

Chronic Exertional Compartment Syndrome



REHAB
REFIT
RETURN TO DUTY

Background

- Chronic exertional compartment syndrome (CECS) causes muscular pain during exercise, mainly in people who run and ruck or march frequently.
- CECS usually develops in the lower leg muscles, but in rare cases it can affect the forearms. It's most common in males and military Service Members.
- Numbness, tingling, weakness, pain, and even foot drop may be felt in either or both legs. In general, this begins within a few minutes of a strenuous activity (for example, running) and usually improves gradually with rest.
- Diagnosis is based on your history and a physical exam. In some cases, intramuscular compartment pressure (IMCP) testing or an MRI may be needed.
- IMCP testing measures fluid pressure inside the muscle compartment. This test must be done before and after exercise.
- The underlying causes of CECS are not fully understood, so treating CECS can be difficult. However, several treatments show good success rates.

Latest Research

- Gait retraining (changing how you run) is often helpful for patients with CECS. It can reduce symptoms in the lower legs and enable a return to running activities. It may take several visits over a few months to fully change your gait pattern.
- Botulinum injections (commonly known as BOTOX) into the affected muscle compartments is an emerging treatment option. Early research results are promising, but research to learn more about the effectiveness of this treatment is ongoing.

Practical Tips for Success

- Temporarily cut back on the amount of the activity that causes pain—both in volume (mileage) and intensity.
- Barefoot running for shorter distances at slow speeds can help improve running form.
- Cross-training, such as cycling, swimming, or rowing, can help maintain strength and aerobic fitness while reducing strain on your lower leg muscles.
- Work hard on rehabilitation to improve gait, strength, balance, and function.

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Rehab Instructions

Gait Retraining

Schedule an appointment with the Fort Belvoir Sports Medicine Clinic for gait analysis.

- By optimizing your running technique, you can often reduce demand on overloaded leg muscles.

Ankle exercises with exercise band (A-D) and Stretches (E)

Ankle Inversion (A) | 🗒 Complete 3 sets of 12 reps 1x a day for 4–12 weeks.




- Loop a resistance band around a secure, stable piece of furniture, and place your foot inside while sitting on the floor or chair.
- Flex only at your ankle and move your foot inward against the band.
- *Slowly* relax your ankle. Don't let the band pull your foot back quickly.

Ankle Eversion (B) | 🗒 Complete 3 sets of 12 reps 1x a day for 4–12 weeks.




- Keep your foot in the loop, and move your body so the band is around the outside of your foot.
- Flex only at your ankle and move your foot outward against the band.
- *Slowly* relax your ankle. Don't let the band pull your foot back quickly.

Ankle Dorsiflexion (C) | 🗒 Complete 3 sets of 12 reps 1x a day for 4–12 weeks.




- Keep your foot in the loop, and move your body so the band is around the outside of your foot.
- Flex only at your ankle and move your foot outward against the band.
- *Slowly* relax your ankle. Don't let the band pull your foot back quickly.

Towel Scrunch (D)
🗒 Complete 2 sets of 10 reps 1x a day for 4–12 weeks.



- Sit in a chair with one foot resting on a towel.
- Spread your toes out and curl them in to scrunch up and draw the towel toward you.
- Using a thicker towel will make it easier to grab with your toes.

Calf Stretch (E)
🗒 Complete 3 sets of 10 reps 1x a day for 4–12 weeks.




- Stand against a wall and place one leg forward. Make sure your knee doesn't go past your toes.
- Stretch the back leg with that knee straight (Stretch 1, left)
- Repeat with your knee bent (Stretch 2, right).
- Hold for 30–60 seconds.